MAKE MY MARRIAGE

Database Design

MongoDB data model, relationships, indexes, and storage boundaries

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Document Guide

This document is organized into the following sections:

1. Modeling Principles
2. Collection Catalog
3. Collection Schemas
4. Indexes and Constraints
5. Relationships
6. Data Isolation and Lifecycle

1. Modeling Principles

- MongoDB stores structured application data and media metadata only.
- Use references for large/growing relationships such as guests, events, tasks, expenses and media.
- Embed small bounded configuration objects such as venue and wedding website settings.
- Use weddingId on almost every wedding-owned collection.
- Store raw public access tokens only on the client; persist secure hashes where practical.

2. Collection Catalog

```
users
weddings
wedding_members
member_invites
events
guests
event_guests
rsvp_tokens
tasks
expenses
shopping_items
vendors
wedding_vendors
event_vendors
invitations
albums
media
upload_sessions
share_links
livestreams
activity_logs
```

3. Users - users

Field	Type	Purpose
_id	ObjectId	Primary key
googleSubjectId	string	Unique Google subject identifier
email	string	Unique normalized email
name	string	Display name
profilePhotoUrl	string|null	Optional avatar
status	ACTIVE|DISABLED	Account state
lastLoginAt	Date	Last successful login
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: UNIQUE googleSubjectId; UNIQUE email.

4. Weddings - weddings

Field	Type	Purpose
_id	ObjectId	Primary key
brideName	string	Bride name
groomName	string	Groom name
weddingDate	Date	Primary wedding date
city	string	Wedding city
state	string|null	State
country	string	Defaults to India

timezone	string	Defaults to Asia/Kolkata
slug	string	Unique public URL slug
coverPhotoMediaId	ObjectId|null	Cover media reference
description	string|null	Optional description
hashtag	string|null	Wedding hashtag
status	DRAFT|ACTIVE|COMPLETED|ARCHIVED	Wedding state
guestUploadsEnabled	boolean	Guest upload master switch
websiteConfig	object	Classic theme and public section settings
createdBy	ObjectId	Creator user
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: UNIQUE slug; INDEX weddingDate; INDEX createdBy; INDEX status.

5. Wedding Members - wedding_members

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
userId	ObjectId	User reference
role	ADMIN|MEMBER	MVP role
status	ACTIVE|REMOVED	Membership state
joinedAt	Date	Join time
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: UNIQUE weddingId + userId; INDEX weddingId + status; INDEX userId + status.

6. Member Invites - member_invites

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
email	string	Invited email
role	MEMBER	Initial role
invitedBy	ObjectId	Inviter user
tokenHash	string	Hash of invite token
status	PENDING|ACCEPTED|EXPIRED| REVOKED	Invite state
expiresAt	Date	Expiry
acceptedAt	Date|null	Acceptance time
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: UNIQUE tokenHash; INDEX weddingId + email; INDEX expiresAt.

7. Events - events

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
name	string	Display name
type	HALDI|MEHNDI|SANGEET|WEDDING| RECEPTION|CUSTOM	Event type
isDefault	boolean	Whether auto-created
description	string|null	Optional description
date	Date	Event date

startTime / endTime	string|null	Local wall-clock time
venue	object	Venue name, address and optional coordinates
dressCode	string|null	Dress code
coverPhotoMediaId	ObjectId|null	Cover media
sortOrder	number	Display order
status	ACTIVE|CANCELLED|ARCHIVED	Event state
createdBy	ObjectId	Creator user
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + date; INDEX weddingId + status; INDEX weddingId + sortOrder.

8. Guests - guests

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
name	string	Guest name
phone	string|null	Optional phone
email	string|null	Optional email
side	BRIDE|GROOM|COMMON	Wedding side
groupName	string|null	Family/group label
city	string|null	Guest city
notes	string|null	Private notes
status	ACTIVE|ARCHIVED	Guest state
createdBy	ObjectId	Creator user
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + name; INDEX weddingId + email; INDEX weddingId + phone; INDEX weddingId + side; INDEX weddingId + groupName.

9. Event Guests - event_guests

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
eventId	ObjectId	Event reference
guestId	ObjectId	Guest reference
invitationStatus	NOT_SENT|SENT	Invitation state
rsvpStatus	PENDING|ATTENDING|NOT_ATTENDING	Event-level RSVP
rsvpUpdatedAt	Date|null	Last RSVP update
notes	string|null	Optional notes
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: UNIQUE eventId + guestId; INDEX weddingId + eventId; INDEX weddingId + guestId; INDEX eventId + rsvpStatus.

10. RSVP Tokens - rsvp_tokens

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
guestId	ObjectId	Guest reference
tokenHash	string	Hash of public token

status	ACTIVE|REVOKED|EXPIRED	Token state
expiresAt	Date|null	Optional expiry
lastAccessedAt	Date|null	Access audit
createdAt	Date	Creation time

Recommended indexes: UNIQUE tokenHash; INDEX weddingId + guestId.

11. Tasks - tasks

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
eventId	ObjectId|null	Optional event scope
title	string	Task title
description	string|null	Description
assignedToUserId	ObjectId|null	Member assignee
priority	LOW|MEDIUM|HIGH	Priority
status	TODO|IN_PROGRESS|COMPLETED	Task state
dueDate	Date|null	Due date
completedAt	Date|null	Completion time
createdBy	ObjectId	Creator user
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + status; INDEX weddingId + eventId; INDEX weddingId + assignedToUserId; INDEX weddingId + dueDate.

12. Expenses - expenses

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
eventId	ObjectId|null	Optional event
weddingVendorId	ObjectId|null	Optional vendor
title	string	Expense title
category	enum	Venue, catering, photography, decoration, clothing, jewelry, etc.
amount	number	Total amount
paidAmount	number	Paid amount
currency	string	INR by default
paymentStatus	PENDING|PARTIALLY_PAID|PAID	Payment state
paymentDate	Date|null	Payment date
receiptMediaId	ObjectId|null	Receipt media
notes	string|null	Notes
createdBy	ObjectId	Creator
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + category; INDEX weddingId + eventId; INDEX weddingId + paymentStatus; INDEX weddingId + weddingVendorId.

13. Shopping Items - shopping_items

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant

eventId	ObjectId|null	Optional event
item	string	Shopping item
category	string|null	Category
forWhom	string|null	Person
assignedToUserId	ObjectId|null	Member owner
storeOrVendor	string|null	Store/vendor text
cost	number|null	Cost
currency	string	INR
status	NEED_TO_BUY|SHORTLISTED| ORDERED|PURCHASED	Shopping state
notes	string|null	Notes
linkedExpenseId	ObjectId|null	Optional linked expense
createdBy	ObjectId	Creator
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + status; INDEX weddingId + eventId.

14. Vendors - vendors

Field	Type	Purpose
_id	ObjectId	Primary key
name	string	Vendor name
category	enum	Vendor category
source	PLATFORM|MANUAL|EXTERNAL	Origin
externalSource / externalId	string|null	Provider identity
phone / email / website	string|null	Contact information
address	object	City/state/country
rating	number|null	Optional external/platform rating
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX category + address.city; INDEX source + externalId.

15. Wedding Vendors - wedding_vendors

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
vendorId	ObjectId|null	Optional generic vendor ref
displayName	string	Wedding-specific display name
category	string	Category
contactPerson	string|null	Contact
phone / email	string|null	Wedding-specific contact
agreedPrice / paidAmount	number|null	Commercial tracking
currency	string	INR
status	SHORTLISTED|CONFIRMED| COMPLETED|CANCELLED	Relationship state
notes	string|null	Private notes
createdBy	ObjectId	Creator
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + category; INDEX weddingId + status.

16. Event Vendors - event_vendors

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
eventId	ObjectId	Event reference
weddingVendorId	ObjectId	Saved wedding vendor
serviceDescription	string|null	Event-specific service
createdAt	Date	Creation time

Recommended indexes: UNIQUE eventId + weddingVendorId; INDEX weddingId + eventId.

17. Invitations - invitations

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
title	string	Invitation title
message	string|null	Message
coverMediaId	ObjectId|null	Cover media
eventIds	ObjectId[]	Included events
rsvpEnabled	boolean	RSVP action
websiteLinkEnabled	boolean	Wedding website action
status	DRAFT|PUBLISHED	Publication state
createdBy	ObjectId	Creator
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + status.

18. Albums - albums

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
eventId	ObjectId|null	Optional event
name	string	Album name
coverMediaId	ObjectId|null	Cover
visibility	PRIVATE|SHARE_LINK	Visibility state
createdBy	ObjectId	Creator
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + eventId.

19. Media - media

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
eventId / albumId	ObjectId|null	Event and album refs
mediaType	IMAGE|VIDEO	Media kind
originalFileName / mimeType	string	File metadata
fileSize	number	Bytes

storageKey	string	S3 original key
thumbnailKey / optimizedKey	string|null	Derived keys
width / height	number|null	Image dimensions
uploaderType	MEMBER|GUEST	Uploader class
uploadedByUserId	ObjectId|null	Member uploader
uploadSessionId	ObjectId|null	Guest/batch session
visibilityStatus	ACTIVE|HIDDEN|DELETED	Media visibility
processingStatus	UPLOADING|PROCESSING|READY|FAILED	Pipeline state
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + eventId; INDEX weddingId + albumId; INDEX weddingId + createdAt; INDEX processingStatus.

20. Upload Sessions - upload_sessions

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
eventId	ObjectId|null	Optional event lock
tokenHash	string	Public token hash
type	GUEST_UPLOAD|MEMBER_UPLOAD	Session type
status	ACTIVE|DISABLED|EXPIRED	State
maxFiles	number	Upload count limit
maxFileSizeBytes	number	Per-file size limit
allowedMimeTypes	string[]	Allowed types
expiresAt	Date|null	Optional expiry
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: UNIQUE tokenHash; INDEX weddingId + status.

21. Share Links - share_links

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
resourceType	ALBUM|GALLERY	Shared resource
resourceId	ObjectId|null	Resource reference
tokenHash	string	Public token hash
status	ACTIVE|REVOKED|EXPIRED	State
expiresAt	Date|null	Optional expiry
allowDownload	boolean	Download permission
createdBy	ObjectId	Creator
createdAt	Date	Creation time

Recommended indexes: UNIQUE tokenHash; INDEX weddingId + status.

22. Livestreams - livestreams

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
eventId	ObjectId	Event reference
title	string	Display title

provider	YOUTUBE|FACEBOOK|INSTAGRAM|OTHER	Provider
streamUrl	string	External stream URL
startTime / endTime	Date|null	Schedule
status	SCHEDULED|LIVE|ENDED	State
createdBy	ObjectId	Creator
createdAt / updatedAt	Date	Audit timestamps

Recommended indexes: INDEX weddingId + eventId; INDEX status + startTime.

23. Activity Logs - activity_logs

Field	Type	Purpose
_id	ObjectId	Primary key
weddingId	ObjectId	Wedding tenant
userId	ObjectId|null	Actor
action	string	Action code
entityType	string	Entity class
entityId	ObjectId|null	Entity reference
metadata	object	Small display/audit metadata
createdAt	Date	Timestamp

Recommended indexes: INDEX weddingId + createdAt; INDEX weddingId + userId.

24. Core Relationship Model

```
User -> WeddingMember -> Wedding
Wedding -> Events
Wedding -> Guests -> EventGuests -> Events
Guest -> RSVPToken
Wedding -> Tasks / Expenses / ShoppingItems
Vendor -> WeddingVendor -> EventVendor -> Event
Wedding -> Invitations
Wedding -> Albums -> Media -> S3
Wedding -> UploadSessions / ShareLinks / Livestreams / ActivityLogs
```

25. Data Isolation Rules

- Every private lookup by resource ID must resolve the resource weddingId and validate membership.
- Do not accept client-provided weddingId as proof of ownership.
- Unique relationships such as eventId + guestId prevent accidental duplication.
- Public token lookups should return only explicitly safe fields.

26. Token Storage and Expiry

- Generate cryptographically secure random tokens.
- Store a SHA-256 or equivalent one-way hash rather than the raw token where practical.
- Allow explicit revocation.
- Support expiry for member invites and optional expiry for RSVP, gallery, and upload links.
- Rate limit token validation endpoints.

27. Media Lifecycle

```
UPLOADING -> PROCESSING -> READY
                        |
                        -> FAILED

Visibility: ACTIVE -> HIDDEN -> DELETED
```

Deleting a media record should use a controlled lifecycle so asynchronous deletion of derived S3 objects cannot leave inconsistent metadata.

28. Money Representation

For the MVP expense tracker, integer INR values are acceptable. If payment processing or multi-currency accounting is added, migrate to smallest currency units and explicit precision rules.

29. Future Schema Extensions

- Advanced role/permission collections.
- Multiple weddings per user.
- Wedding planner organizations.
- Accommodation and transportation.
- Seating plans.
- Vendor quotes and payments.
- WhatsApp communication audit.
- AI-generated checklists and recommendations.